

BRIEF COMMUNICATION

**MATURE BEECH TREES (*FAGUS GRANDIFOLIA*; FAGACEAE) ARE
PERSISTENTLY CLONAL IN COVES AND BEECH GAPS IN THE
GREAT SMOKY MOUNTAINS¹**

ASHLEY B. MORRIS^{2,5}, NICOLE K. MILLSAPS², KELSIE Q. MASON², JACQUELYN S. HOWELL³,
AND JAMES H. SPEER⁴

²Department of Biology, Middle Tennessee State University, Murfreesboro, Tennessee 37132 USA; ³Department of Biology, University of South Alabama, Mobile, Alabama 36688 USA; and ⁴Department of Earth and Environmental Systems, Indiana State University, Terre Haute, Indiana 47809 USA

- *Premise of the study:* Angiosperms frequently have mixed sexual and asexual reproductive strategies, which can have significant consequences for population and community structure. Many tree species respond to disturbance by vegetative sprouting over sexual reproduction, but the ability to do so varies within and among species and is poorly studied. We tested the hypothesis that root sprouting in *Fagus grandifolia* is more important in high-elevation beech forests (extreme environmental conditions), relative to lower-elevation cove hardwood forests (optimal environmental conditions), in Great Smoky Mountains National Park.
- *Methods:* Four cove hardwood and two high-elevation beech gap forests were sampled. *Fagus grandifolia* trees were tagged, mapped, and genotyped at each site using six nuclear microsatellite loci. A subset of trees was cored to determine minimum ages of identified clonal lineages. Relationships between measures of clonal richness, diameter, forest type, and elevation were compared by nonparametric analyses.
- *Key results:* Five of six sites were highly dependent on sprouting; one site was entirely dependent on reproduction by seed. Clonal richness did not differ significantly by forest type or elevation, but differed significantly from previously published work. Tree diameter was significant by elevation and significant between the present study and previously published work.
- *Conclusions:* The results of this study in the context of previously published work suggest that relatively recent recruits may be more likely to reflect both sexual and asexual strategies, whereas mature *F. grandifolia* may be more likely to be the result of persistent clonal lineages.

Key words: American beech; clonal reproduction; Fagaceae; microsatellites; root sprouting.

Flowering plants commonly exhibit a mixed reproductive strategy in which both sex and clonal growth contribute to population structure. Clonality, also known as vegetative growth, has been described as a response to sexual failure and is typical of less favorable habitats (van Groenendael et al., 1997; Silvertown, 2008). Questions remain, however, as to what biotic or abiotic conditions trigger clonal reproduction in species that exhibit both sexual and asexual strategies across their geographic ranges. Understanding the drivers of plant reproductive strategy has broad-reaching significance for ecologists and evolutionary biologists alike. Ecological modelers are building increasingly more sophisticated species distribution models (SDMs) to forecast potential future distributions under climate change, as well

as to hindcast paleodistributions for phylogeographic studies. These models are just now beginning to integrate measures beyond typical environmental layers, such as shade tolerance, seedling establishment, susceptibility to drought, and others (Matthews et al., 2011). Given that clonal architecture and spatial genetic structure within populations can have significant implications for mating patterns (Vallejo-Marin et al., 2010), reproductive strategy should be considered an important indicator of migration potential for such models. Additionally, reproductive strategy may play a role in speciation rates and extinction and should therefore be further explored across phylogenetic space (Holsinger, 2000). These represent just a few examples of why understanding the mechanisms that drive reproductive strategy within species is of broad importance to the botanical community.

Many woody plants, particularly trees, can reproduce asexually through the production of secondary trunks, commonly referred to as sprouting. This vegetative sprouting may occur via collar sprouts, specialized underground stems, or root sprouts (Del Tredici, 2001). Sprouting ability varies with age and by site, yet this variation has been poorly characterized within most species (Bond and Midgley, 2001). Sprouting behavior has implications for forest community dynamics, such that understanding this strategy on local and broader scales will facilitate our ability to model host–pathogen interactions, species

¹Manuscript received 30 April 2013; revision accepted 6 January 2014.

The authors thank Glen Taylor and Keith Langdon at GRSM for assistance with site selection and securing collection permits. The authors thank Greg Wiggins, Chris Morris, and Nick Morris for extensive field assistance. The authors thank Chris Morris for assistance with statistical analyses. Finally, they thank Sarah Bergemann, Michael Held, Matt Gitzendanner, the Associate Editor, and two anonymous reviewers for constructive comments on a previous version of this manuscript.

⁵Author for correspondence (e-mail: ashley.morris@mtsu.edu)

responses to climate change, and to develop forest management practices otherwise limited by the lack of such data (Bond and Midgley, 2003). In the present study, we used *Fagus grandifolia* Ehrh. (American beech, Fagaceae) as a case study to investigate factors regulating a system in which both sexual and asexual reproduction (via root sprouting) occur.

Fagus grandifolia is one of several widespread eastern North American tree species that is predicted to experience a significant range reduction under climate change (Iverson et al., 2008). This species also faces the continuing threat of beech bark disease (BBD), a scale insect (*Cryptococcus fagisuga* Lind.) and *Neonectria* (*N. ditissima* Samuels & Rossman or *N. faginata* Castlebury) fungal complex, which has altered beech-dominated forests throughout the northern portions of eastern North America over the last 100 yr (Koch, 2010; Cale et al., 2012; Mason et al., 2013). As the geographic area affected by BBD continues to expand, understanding the relationships between the host and pathogen will be vital to the effective management of outbreaks. Our ability to accurately predict how *F. grandifolia* will respond to the combined threats of climate change and disease is largely limited by our current understanding of host ecology. *Fagus grandifolia* is a prolific root sprouter (Tubbs and Houston, 1990), and the frequency of root sprouting varies among populations (Kitamura and Kawano, 2001). However, the factors dictating the propagation or spread of populations through asexual and sexual reproduction remain elusive (Morris et al., 2004). Clarifying the relationship between root sprouting, environmental conditions, and evolutionary history, will improve the robustness of models used to predict species responses to climate change (Beaudet and Messier, 2008; Takahashi et al., 2010; Matthews et al., 2011).

The purpose of our study was to assess clonal structure in mature *F. grandifolia* on a local scale in Great Smoky Mountains National Park (GRSM), where the species occurs across a broad elevational and ecological range. Whittaker (1956) described *F. grandifolia* in GRSM as being most common in mesic, nonquercine forests. These forests can be further subdivided into cove hardwood, eastern hemlock, and gray beech or beech gap forests; *F. grandifolia* is relatively rare in the eastern hemlock forest. Previous studies in GRSM reached conflicting conclusions regarding the role of root sprouting in cove hardwoods and beech gaps (Kitamura et al., 2001; Morris et al., 2004). Kitamura et al. (2001) used allozymes to sample one cove hardwood and one beech gap site and determined both to be dependent on root sprouting. Morris et al. (2004) used inter-simple sequence repeats (ISSRs) to sample three cove hardwood and two beech gap sites and determined all but one to be dependent on sexual reproduction. Because of differences in field sampling strategies between the two studies, direct comparisons are difficult to make. However, given the significance of root sprouting as a functional trait (Bond and Midgley, 2001, 2003), a more thorough understanding of this system is warranted.

In the present study, we specifically evaluated the hypothesis that dependence on root sprouting should be greater in beech gap forests than in cove hardwood forests due to increasingly harsher winter conditions experienced at higher elevations in GRSM. As described by Whittaker (1956) and others, cove forests in GRSM represent optimal growing conditions for *F. grandifolia*, typically having more loamy soils with mean annual temperatures ranging from 4°–27°C. Alternatively, the beech gap forests represent more extreme environmental conditions for *F. grandifolia*, with thin, rocky soils and mean annual temperatures ranging

from 6°–10°C cooler than cove forests. While these two forest types are not explicitly defined by elevation, Whittaker's descriptions suggest that they meet between 1280 and 1370 m above sea level (a.s.l.) Russell (1953) defined beech gaps as beech-dominated, deciduous islands surrounded by spruce–fir forests. These stands are suspected to be of root-sprout origin and reportedly most infrequently in response to environmental extremes (Russell, 1953; Whittaker, 1956; Ward, 1961). We tested the hypothesis that high-elevation beech gaps would exhibit greater clonal structure relative to beech in cove hardwood forests. We were particularly interested in mature beech as a measure of historical reproductive strategy in these forests, rather than recent recruitment in the context of BBD-induced disturbance. We used six nuclear microsatellite loci to assess this hypothesis in *F. grandifolia* in GRSM, and we compared the results of the present study to that of Morris et al. (2004).

MATERIALS AND METHODS

Field sampling—Six sites were chosen for sampling in GRSM (Fig. 1): four cove hardwood sites and two beech gap sites. The six sites, by increasing elevation above sea level, were Grassy Branch (GRB; 466 m a.s.l.), Cataloochee (CAT; 711 m), Heintooga Road (HR; 1127 m), Chimney Tops Trail (CHM; 1147 m), Balsam Mountain Road (BMT; 1488 m), and Jenkins Knob (JK; 1698 m). The CHM and JK sites are permanent monitoring plots for BBD progression, first established in 1994 (Wiggins et al., 2004). For the purposes of this study, GRB, CAT, HR, and CHM are cove hardwood sites, and BMT and JK are high-elevation beech gap sites (Fig. 1).

All sites were sampled in June 2001 and June 2002. Within each site, latitude, longitude, and elevation were recorded using a Garmin Etrex Global Positioning System (GPS), aspect was recorded, associated shrubs and trees were recorded, and canopy cover was photo-documented using a Canon EOS Rebel with a fisheye lens. Three images were taken in each site: one in plot center, one on the north side of the plot, and one on the south side of the plot. Images were analyzed using the software program Gap Light Analyzer (GLA) v. 2.0 (Frazer et al., 1999). Site characteristics for the six sites sampled in this study are summarized in Table 1. Sites were a mix of north- and south-facing, with percentage canopy cover relatively consistent across sites and elevations, although beech gaps exhibited slightly more open canopy (20–29%), compared to cove sites (19–25%). The incidence of BBD was noted in five of the six sites, with trees in the GRB site lacking any symptomatic bark tissues.

A maximum of 20 trees of *F. grandifolia* (American beech, or here simply “beech”) were initially mapped and sampled at each site, for a total of 120 trees across sites. Within each site, one tree was chosen as the plot center based on its central position in the stand relative to extent observed for the species. From the plot center, nearest neighbors were sampled in a radial fashion for a maximum of 20 trees. This approach was taken to standardize the number of trees sampled across sites. Previous work indicated that plots or transects of a predetermined size may yield many trees in beech gaps, while yielding few in cove forests (Morris et al., 2004). Given that root excavation studies suggest that clonal growth in *F. grandifolia* expands radially from the parent tree, with most sprouts occurring within a few meters of the parent tree (Jones and Raynal, 1986), we recognize that our strategy may induce clonal bias. However, because the same strategy was applied across all sites, any such bias should not impact our ability to make inferences between cove and beech gap sites.

Only trees of 3 cm DBH or larger were sampled to target stems that had arisen before the arrival of BBD in GRSM in the early 1990s. BBD-aftermath forests have been suggested to be highly clonal in response to high mortality rates of canopy trees, and our strategy was to exclude BBD as a possible factor inducing clonal structure across sites.

Each sampled tree was permanently tagged with numbered aluminum tags nailed to the tree approximately 1.5 m from the base. Trees were mapped relative to the plot center by measuring distance and angle from the center tree. These data were then converted to coordinates and mapped using ArcGIS 10.1 (ESRI, Redlands, California, USA). For each sampled tree, DBH was measured, and a leaf was collected and preserved in silica gel desiccant for genetic analysis. Vouchers for four of the six sites were also collected and are housed at the Middle Tennessee State University Herbarium (MTSU; Appendix 1). While there were no vouchers collected for the two remaining sites, all trees

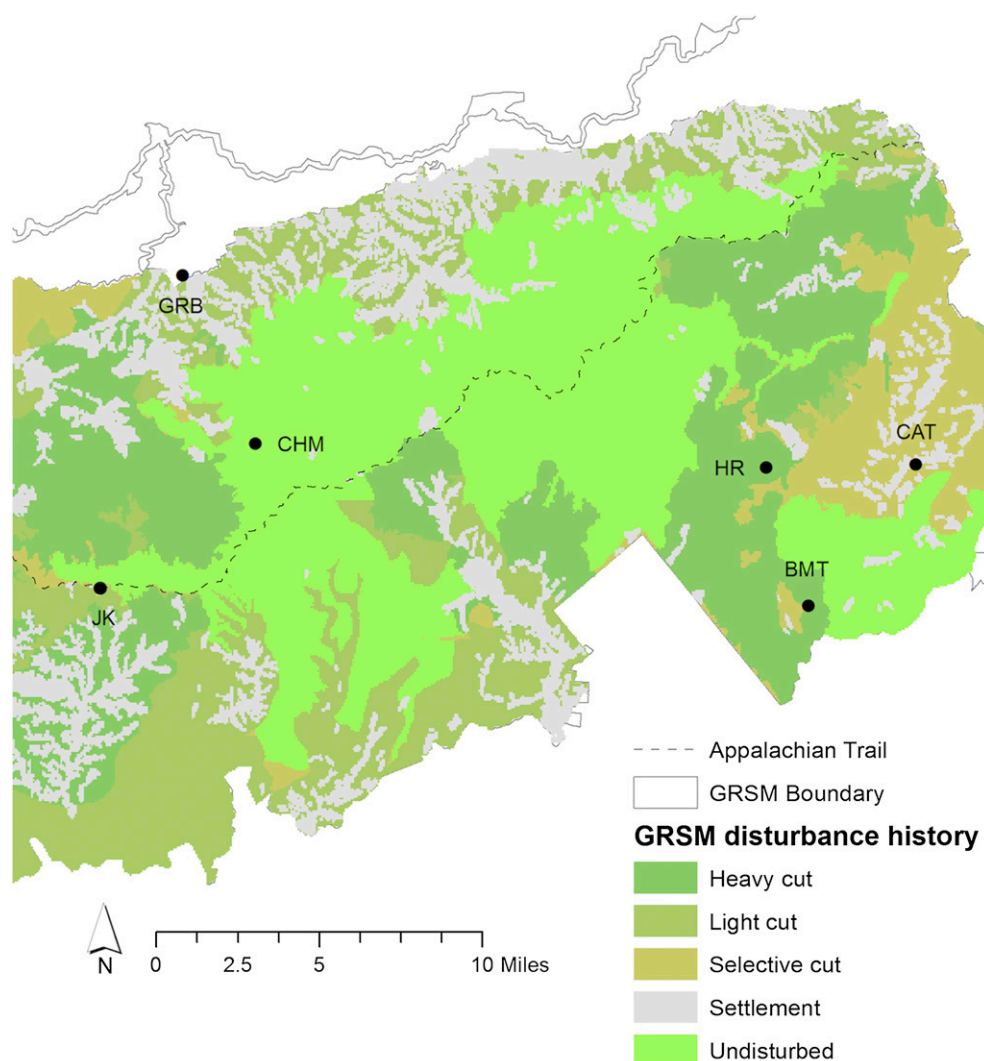


Fig. 1. Stands of *Fagus grandifolia* sampled in Great Smoky Mountains National Park, United States, for the present study. The six sites, by increasing elevation, were Grassy Branch (GRB; 466 m a.s.l.), Cataloochee (CAT; 711 m), Heintooga Road (HR; 1127 m), Chimney Tops Trail (CHM; 1147 m), Balsam Mountain Road (BMT; 1488 m), and Jenkins Knob (JK; 1698 m). Sites CHM and JK are permanent plots, first established in 1994, to monitor BBD progression (Wiggins et al., 2004). For the purposes of this study, GRB and CAT are low-elevation sites, HR and CHM are mid-elevation sites, and BMT and JK are high-elevation sites. Sampled sites are in the context of historical disturbance as recorded by Pyle (1988). Site characteristics are given in Table 1; clonal structure by site as determined by microsatellites is given in Table 3.

sampled in the present study were permanently tagged and mapped, such that each tree can be relocated in the field if other researchers require additional information.

In June 2002, sampling was expanded to include additional trees at each site to increase sample sizes, with the exception of the HR site. The HR site is in a

topographically variable region of the park, and the GPS unit was insufficiently accurate to relocate the site after multiple attempts. For the five remaining sites, 10 additional trees were sampled according to the strategy used in 2001, resulting in a total of 30 trees per site (20 trees for HR) and 170 trees across all six sites.

TABLE 1. Field site characteristics for *Fagus grandifolia* stands sampled in Great Smoky Mountains National Park for the present study.

Site	Elevation (m a.s.l.)	Aspect	Canopy openness (%) (N / C / S)	Associates
Grassy Branch (GRB)	466	N (0°)	19 / 18 / 22	<i>Acer</i> , <i>Liriodendron</i>
Cataloochee (CAT)	711	SE (140°)	19 / 22 / 19	<i>Aesculus</i> , <i>Carpinus</i> , <i>Oxydendrum</i> , <i>Kalmia</i> , <i>Tsuga</i>
Heintooga Road (HR)	1127	SSE (155°)	19 / 25 / 22	<i>Acer</i> , <i>Liriodendron</i> , <i>Tsuga</i> , <i>Ulmus</i>
Chimney Tops Trail (CHM)	1147	NW (320°)	17 / 17 / 16	<i>Acer</i> , <i>Aesculus</i> , <i>Halesia</i>
Balsam Mountain (BMT)	1488	SSE (145°)	20 / 25 / 29	<i>Abies</i> , <i>Acer</i> , <i>Picea</i> , <i>Ulmus</i>
Jenkins Knob (JK)	1698	NE (45°)	21 / 25 / 26	<i>Aesculus</i> , <i>Picea</i>

Notes: Canopy openness was calculated as a percentage based on photos taken at north, center, and south sides of each site and analyzed with Gap Light Analyzer (GLA) v. 2.0 (Frazer et al., 1999).

Tree core analysis—Dendrochronological analyses were performed to provide minimum ages for any detected clones within the sampled sites. Three of the 20 trees sampled at each site in 2001 were cored 1.5 m above ground, and the cores were analyzed to determine the ages of these trees. Sampled trees were selected to reflect the range of diameters observed at each site. Regression analyses were used to determine the degree of correlation between tree diameter and age. This test was performed to determine whether tree age could be confidently extrapolated from diameter at each site.

Microsatellite genotyping—Genomic DNA was extracted from silica-dried leaves using the Qiagen DNeasy Plant Mini Kit (Qiagen, Madison, Wisconsin, USA). Ten nuclear microsatellite loci previously developed from *F. grandifolia* EST libraries (Kubisiak et al., 2009) were screened for utility in this project. Due to problems with amplification or amplification profiles, only six of these loci were used here: *FgSI0006*, *FgSI0009*, *FgSI0015*, *FgSI0016*, *FgSI0017*, and *FgSI0026*. Amplification conditions followed those described in Kubisiak et al. (2009). Loci were individually amplified and then up to three loci were multiplexed and run on an ABI 3130XL DNA analyzer (Applied Biosystems, Life Technologies, Grand Island, NY) with 500 LIZ size standard. Data were scored with GeneMarker MTP software (SoftGenetics, State College, Pennsylvania, USA), and all calls were manually confirmed.

Clonal assignment, clonal structure, and genetic diversity—To assess genotypic resolution of the loci in the present study, we used a resampling procedure as implemented in the software program GenClone v. 2.0 (Arnaud-Haond and Belkhir, 2007). Resampling all loci across all sampled units as implemented in GenClone v. 2.0 (Arnaud-Haond and Belkhir, 2007) provides box plot data for all possible combinations (C) of $c \frac{K}{l}$ of K loci ($K = 1, \dots, l$, where l is the number of loci available; Arnaud-Haond and Belkhir, 2007). The bottom and top lines of each box indicate the minimum and maximum number of genotypes, while the central line within each box shows the average number of genotypes generated by each combination of l loci. An asymptotic curve indicates that the loci observed provide an accurate determination of the number of genotypes in the sample. The resulting box plots illustrate the number of multilocus genotypes (MLGs) recovered from all possible combinations of $n - 1$ loci.

Clonal assignment was determined using a 1-step or a 2-step threshold under the stepwise mutation model as implemented in the software program GenoDive 2.0b23 (Meirmans and Van Tienderen, 2004). Only individuals with complete data at all loci were included in the analysis to prevent missing data from unduly influencing the assignment to a genotype. This assessment was done to account for the potential for genotyping error or somatic mutation. A 1-step threshold allows two trees to differ by a single allele at a single locus and still be considered clonemates, while a 2-step threshold would allow clonemates to differ by 2-mutational steps. The 1- and 2-step thresholds each recovered 46 MLGs, and no MLGs were shared across sites. Therefore, the 1-step and 2-step thresholds were equivalent, and the 2-step threshold was selected for all other clonal diversity analyses (Appendix S1, see Supplemental Data with the online version of this article). Next, we tested the assumption that any shared genotypes were a consequence of clonality, rather than being a function of sexual reproduction, using an approach by Gómez and Carvalho (2000) implemented in GenoDive 2.0b23. The test determines whether genotypic diversity in a sample size of size N is consistent with expectations for a sexual, panmictic population under Hardy–Weinberg equilibrium.

Clonal structure was characterized by a series of measures as recommended by Arnaud-Haond and Belkhir (2007) and implemented in GenoDive 2.0b17 (Meirmans and Van Tienderen, 2004). Clonal richness (R) was calculated as $(G - 1) / (N - 1)$ where G is the number of genotypes and N is the number of individuals sampled. R is a modification of proportion distinguishable (PD; Ellstrand and Roose, 1987) that accounts for small sample sizes (Dorken and Eckert, 2001). Simpson's complement (D^* ; Pielou, 1969) was calculated as $1 - \sum_i (n_i - 1) / (N(N - 1))$ where n_i is the number of samples with genotype i . The statistic D^* indicates the probability that two randomly selected individuals from the sample will each represent unique multilocus genotypes (MLGs). The largest clonal lineage (LCL), based on number of individuals and LCL patch size, was also estimated.

Microsatellite loci summary statistics (number of observed alleles, size range of observed alleles, and observed and expected heterozygosity) and tests for deviation from Hardy–Weinberg equilibrium were calculated with GenoDive 2.0b23 after collapsing the data set to single replicates of unique MLGs.

Statistical analysis—A series of nonparametric tests were performed to determine any potential correlations between clonal richness (R) and forest type

(cove vs. gap) or elevation (low, mid, and high), as well as any correlations between DBH and forest type or elevation. Data reported by Morris et al. (2004) were used to calculate R values for the three cove and two beech gap sites reported in that study. Previously unpublished DBH data for the sites of Morris et al. (2004) were recovered for the present analysis. All analyses were first performed on the data from the present study only, followed by analyses on a combined present and 2004 data set, and then on comparisons between the present and past studies. Nonparametric comparisons of DBH and R values were accomplished with Wilcoxon's rank sum test when there were two comparison groups and the Kruskal–Wallis test when there were three comparison groups.

RESULTS

Tree core analysis—Three trees were cored at each of the six sites, for a total of 18 trees. Trees ranged in age from 34 to 156 yr, with the average age of 73 yr old (Table 2). The age–diameter relationship was significant ($P < 0.001$) with age explaining 46% of the variance in diameter.

Microsatellite genotyping—Of 170 trees sampled for this study, 150 were successfully genotyped for all six loci. The remaining 20 individuals either failed in all amplifications or were only successfully amplified for a fraction of loci after repeated attempts. Additionally, subsequent attempts were made to resample “problem” trees in the field. However, several of these trees had died between sampling periods as a result of BBD. Missing data (defined as one or more loci for which an individual failed to amplify after multiple attempts or for which an individual consistently produced a poor profile) were removed before the analyses. The frequency of alleles observed across six loci was consistent with those previously reported by Kubisiak et al. (2009), with the exception of two novel alleles at locus *Fg0016* (Table 3). Observed size ranges varied from those reported in Kubisiak et al. (2009), in which 24 trees from a single stand in Michigan, United States, were surveyed. Additional work is underway to determine whether this is the result of genotyping on different sequencers or if it reflects a broader range of allelic variation for this widespread species.

Clonal assignment, clonal structure, and genetic diversity—Based on a resampling procedure as implemented in GenClone v. 2.0 (Arnaud-Haond and Belkhir, 2007), genetic resolution was sufficient to adequately determine individual MLGs for this data set (Appendix S2; see online Supplemental Data). The test for probability of finding the observed clonal diversity under random mating was significant for all sites ($P = 0.001$) except for GRB ($P = 1.00$). Clonal richness (R) under the 2-step threshold varied from 0.00 at the highest elevation site (JK) to 1.00 at the lowest elevation site (GRB) (Table 2). The largest clonal lineage (LCL) was recovered at JK, with 27 trees sharing an identical MLG, and a minimum age of 76 yr based on tree core analysis. The sizes and ages of the LCLs varied at the remaining sites exhibiting sprouting, with the oldest minimum LCL recovered at CHM, with an estimated minimum age of 152 yr. Patch sizes covered by LCLs also varied by site (Fig. 2), with BMT exhibiting the largest patch size at 1125 m².

Statistical analysis—The results of statistical analyses are summarized in Table 4 and Fig. 3. For the present study, R values were not significantly different when stratifying by either forest type (cove vs. hardwood; $\chi^2 = 0.21$, $df = 1$, $P = 0.6434$) or elevation (low, mid, or high; $\chi^2 = 0.29$, $df = 2$, $P = 0.8669$). There was also no significant difference found after expanding

TABLE 2. Clonal structure of six stands of *Fagus grandifolia* in Great Smoky Mountains National Park based on six nuclear microsatellite loci.

Site	Elev (m a.s.l.)	N	2-Step			LCL	LCL patch size (m ²)	LCL minimum age
			G	R	D*			
GRB	466	23	23	1.00	1.00	1	—	—
CAT	711	30	5	0.14	0.45	22	240	48
HR	1127	18	4	0.18	0.74	7	50	125
CHM	1147	30	6	0.17	0.69	16	357	152
BMT	1488	22	7	0.29	0.82	8	1125	74
JK	1698	27	1	0.00	0.00	27	70	76

Notes: N = number of trees sampled; G = number of genets, or unique multilocus genotypes, determined by six nuclear microsatellite loci using a threshold of two mutational steps; R = measure of clonal richness, calculated as $(G - 1)/(N - 1)$, provides an estimate of clonal input, in which a monoclonal stand is 0 and a completely sexual stand is 1; D* = measure of clonal richness standardized by resampling, known as Simpson's complement, describes probability of encountering distinct multilocus genotypes in a random sample of two individuals; LCL = largest clonal lineage based on total number of stems; LCL patch size was estimated from mapped individuals. LCL minimum age was based on a tree core analysis of selected individuals within each stand.

the data set to include sites of Morris et al. (2004): forest type ($\chi^2 = 0.14$, df = 1, $P = 0.7048$), elevation ($\chi^2 = 2.00$, df = 2, $P = 0.3732$). Overall differences in R between the Morris et al. (2004) data and the present study were not significant ($\chi^2 = 3.69$, df = 1, $P = 0.0547$). However, the R value for GRB in the present study appears to be a strong outlier, and if it is excluded from the data set of the present study and that of Morris et al. (2004), the R comparison between studies is significant ($\chi^2 = 6.82$, df = 1, $P = 0.0143$).

For the present study, tree DBH did not differ significantly between forest types ($\chi^2 = 0.01$, df = 1, $P = 0.9138$). The median beech gap tree was 10 cm (4.8–18.6 cm); the median cove tree was 9.2 cm (3.0–93.0 cm). There was a significant difference in DBH by elevation ($\chi^2 = 17.99$, df = 1, $P = 0.0001$). The median low-elevation tree was 14.1 cm (3.8–93.0 cm); the median mid-elevation tree was 6.4 cm (3.0–70.4 cm); and the median high-elevation tree was 10 cm (4.8–18.6 cm). The trees in the present study were significantly larger than those of Morris et al. (2004) ($\chi^2 = 128.40$, df = 1, $P < 0.0001$; Fig. 3). The overall median DBH for the present study was 9.9 cm (2.0–93.0 cm) compared to only 1.5 cm (0.10–30.2 cm) in the Morris et al. (2004) study.

DISCUSSION

Mature beech is persistently clonal across forest types—We hypothesized that beech gaps would be more clonal than cove forests in response to the frequency of freezing and thawing cycles experienced in subalpine sites. Previous studies supported

this mechanism as one of several that induce root sprouting in *F. grandifolia* (Held, 1983; Jones and Raynal, 1986, 1988). Furthermore, because of the unique morphology and ecology of high-elevation beech gaps in the southern Appalachian Mountains of eastern North America, several groups have proposed that seed production is limited and population expansion occurs by root sprouting (Russell, 1953; Whittaker, 1956; Ward, 1961). Our results confirm that root sprouting is common in beech gaps, but it is also common in mid- to low-elevation beech forests in GRSM (Fig. 2; Table 2).

Our results confirm previous findings by Kitamura et al. (2001), who found high levels of sprouting in GRSM in both a high-elevation and a low-elevation beech forest based on allozyme data. The one exception to this is GRB, a low-elevation site that we found to be entirely dependent on reproduction by seed ($R = 1.00$; Table 2). However, our results are in contrast to the ISSR-based findings of Morris et al. (2004), who found sprouting to be relatively infrequent in beech forests in GRSM, regardless of elevation. The sampling strategy of Morris et al. (2004) differed from the present work in using transects and rectangular plots and including all size classes in the sampled area. As was previously mentioned, our strategy of sampling nearest neighbors for the present study could potentially bias our results toward identifying greater clonal structure. However, the probability of entering six different sites and consistently sampling clonal lineages at five of those sites as a consequence of sampling strategy alone seems unlikely. Upon comparing DBH between the present study and that of Morris et al. (2004), we found some interesting patterns that might further explain the conflicting results observed here.

TABLE 3. Characterization of six nuclear microsatellite loci used to assess clonal structure in *Fagus grandifolia* in six stands in Great Smoky Mountains National Park, United States.

Locus	Base pairs	N _a	H _o	H _s	H _i	p-value
FgSI0006	256–259	2	0.307	0.425	0.435	0.152
FgSI0009	208–214	3	0.450	0.415	0.437	0.657
FgSI0015	136–139	2	0.283	0.289	0.489	0.685
FgSI0016	146–167	7	0.610	0.599	0.799	0.709
FgSI0017	225–231	3	0.383	0.389	0.418	0.665
FgSI0026	147–156	3	0.398	0.395	0.516	0.299

Notes: Data presented here describe results by locus. Loci are from Kubisiak et al. (2009). bp = Base pairs or allele size range for each locus across sites; N_a = number of alleles observed; H_o = observed heterozygosity; H_s = expected frequency of heterozygotes in subpopulations assuming HWE; H_i = expected frequency of heterozygotes over all populations assuming HWE; p value = one-sided p values of G_{IS}-based estimate of Hardy–Weinberg equilibrium per locus across populations.

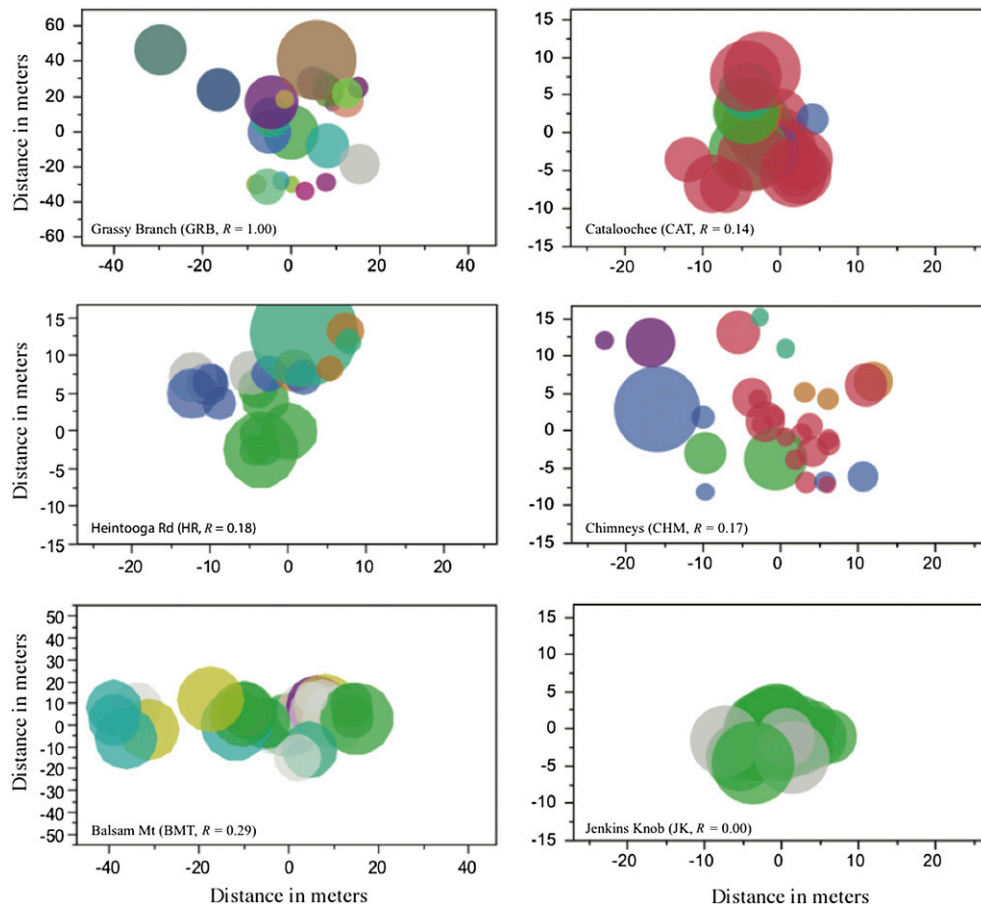


Fig. 2. Clonal structure of *Fagus grandifolia* by site sampled in Great Smoky Mountains National Park. All plots are drawn to the same scale; circles represent individual trees; colors within plots represent unique genotypes; gray circles indicate trees for which no genotype was recorded. No genotypes were shared across sites. Circle size reflects relative tree diameter at breast height, and relative circle size is consistent across plots. The x- and y-axes of each plot are in meters, with the center of each plot being 0, 0. Site abbreviations are consistent with Table 1, and clonal richness values (R) are as described in the text.

Sampled trees in the 2004 study were consistently smaller, and in many cases much smaller, than the trees sampled in the present study (Fig. 3). The 2004 study reported greater dependence on sexual reproduction among sampled stems, while we report greater dependence on asexual reproduction here. This might suggest that smaller size classes (i.e., recent recruits) exhibit a more even distribution of stems of seed vs. root sprout origin. *Fagus grandifolia* can produce copious amounts of nuts by 40 yr of age, with good mast crops occurring at 3- to 8-yr intervals (Tubbs and Houston, 1990). Kitamura et al. (2001, p. 226) indicated that the relative densities of suckers vs. seedlings vary by site and that in at least three of their sites, “offshoot formation by root sucker plays a significant role in stand development and maintenance”. Beaudet and Messier (2008) have shown that root suckers have a competitive advantage over seedlings, such that they have a greater potential of surviving into maturity than do seedlings. However, the dynamics that control the relative survival of seedlings and suckers at any given site may be complex and difficult to unravel.

Caveats and future directions—GRSM represents a somewhat unique environment in which to study the reproductive

ecology of *F. grandifolia*. Few areas provide such a variety of habitats in which to test hypotheses regarding the factors driving the balance between sex and vegetative growth. However, it is also an area rife with potentially confounding factors. Logging occurred throughout the region until GRSM was officially dedicated in 1940 (Campbell, 1969). The minimum age of the largest clonal lineage in four of the six sites sampled in the present study could be a reflection of this period of the area’s history (Table 2). The most complete map to date of logging disturbance in the park is by Pyle (1988) (Fig. 1), but uncertainties remain as to how impacted various stands within the park may have been. More recently, BBD has resulted in high mortality of *F. grandifolia* in the highest elevations of the park, significantly changing the competitive landscape within those communities. Furthermore, some have suggested that ecotypic variants of this species converge in the park, making it difficult to control for the potential role of evolutionary history in reproductive adaptive strategies. These combined factors make it challenging to design a truly balanced study with sufficient replication to verify the conclusions suggested herein. Therefore, future work will expand to include other physiographic regions to further test environmental and abiotic factors that drive reproductive strategy in *Fagus grandifolia*.

TABLE 4. Results of statistical comparisons of clonal richness (R), forest type, elevation, and diameter within and among studies. Significant p -values are in boldface. See text for full explanation of analyses.

Data compared	χ^2 (df)	p
Present study only		
R by forest type (cove vs. beech gap)	0.21 (1)	0.6434
DBH by forest type (cove vs. beech gap)	0.01 (1)	0.9138
R by elevation (low, mid, and high)	0.29 (2)	0.8669
DBH by elevation (low, mid, and high)	17.99 (1)	0.0001
Present study with Morris et al. (2004)		
R by forest type (cove vs. beech gap)	0.14 (1)	0.7048
DBH by forest type (cove vs. beech gap)	2.00 (2)	0.3732
Present study with Morris et al. (2004) (GRB site included / excluded)	3.69 (1) / 6.82 (1)	0.0547 / 0.0143
DBH by study	128.40 (1)	<0.0001

One of the greatest challenges to a study of clonal reproduction is finding the appropriate sampling strategy, particularly for systems in which the extent and pattern of clonal structure are poorly understood. In previous work, Morris et al. (2004) chose to incorporate different sampling strategies under different environments. However, this makes comparisons among sites difficult. In the present study, we chose to use one strategy across all sites, sampling nearest neighbors in a radial fashion. As discussed, this approach also induces potential bias into the data. Here we statistically compared the present data to that of Morris et al. (2004). We fully acknowledge that differences in sampling strategies between the two studies confounds our ability to clearly separate the potential effects of methodology choice from actual differences in reproductive strategy among sites and between size classes. However, our findings do suggest an avenue for future study in the comparison of clonal

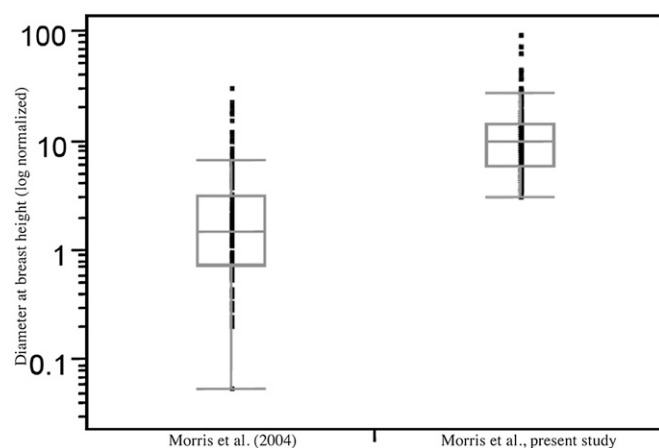


Fig. 3. Diameter of trees of *Fagus grandifolia* sampled in the present study relative to data of Morris et al. (2004). Diameters (on y-axis) have been log-normalized. Study group (2004 on left, present study on right) on x-axis. The two groups differed significantly ($p < 0.0001$); trees of Morris et al. (2004) were significantly smaller than those sampled in the present study.

structure between size classes as a measure of clonal longevity. Finally, it should be noted that sample size can have a significant impact on values of clonal richness, such that the addition or removal of a single sample can change the R value quite a bit. Our experience suggests that sample sizes of 30 or more will reduce the influence of a single sample on such measures. To compare differences in clonal structure between size classes in *F. grandifolia* will require a strategy that samples all individuals of all sizes within a large spatial area. Furthermore, the amount of variation we have observed in clonal structure among sites within a relatively small geographic area suggests that many sites (likely paired sites) will need to be sampled across the full range of the species to fully understand the factors that drive reproductive strategy. Understanding such a complex system will be the first step in improving ecological models relating to population dynamics and competition, as well as to SDMs seeking to predict future responses to climate change.

LITERATURE CITED

- ARNAUD-HAOND, S., AND K. BELKHIR. 2007. GENCLONE: A computer program to analyse genotypic data, test for clonality and describe spatial clonal organization. *Molecular Ecology Notes* 7: 15–17.
- BEAUDET, M., AND C. MESSIER. 2008. Beech regeneration of seed and root sucker origin: A comparison of morphology, growth, survival, and response to defoliation. *Forest Ecology and Management* 255: 3659–3666.
- BOND, W. J., AND J. J. MIDGLEY. 2001. Ecology of sprouting in woody plants: The persistence niche. *Trends in Ecology & Evolution* 16: 45–51.
- BOND, W. J., AND J. J. MIDGLEY. 2003. The evolutionary ecology of sprouting in woody plants. *International Journal of Plant Sciences* 164: S103–S114.
- CALE, J. A., S. K. LETKOWSKI, S. A. TEALE, AND J. D. CASTELLO. 2012. Beech bark disease: An evaluation of the predisposition hypothesis in an aftermath forest. *Forest Pathology* 42: 52–56.
- CAMPBELL, C. C. 1969. Birth of a national park in the Great Smoky Mountains: An unprecedented crusade which created, as a gift of the people, the nation's most popular park. University of Tennessee Press, Knoxville, Tennessee, USA.
- DEL TREDICI, P. 2001. Sprouting in temperate trees: A morphological and ecological review. *Botanical Review* 67: 121–140.
- DORKEN, M. E., AND C. G. ECKERT. 2001. Severely reduced sexual reproduction in northern populations of a clonal plant, *Decodon verticillatus* (Lythraceae). *Journal of Ecology* 89: 339–350.
- ELLSTRAND, N. C., AND M. L. ROOSE. 1987. Patterns of genotypic diversity in clonal plant-species. *American Journal of Botany* 74: 123–131.
- FRAZER, G., C. CANHAM, AND K. LERTZMAN. 1999. Gap Light Analyzer (GLA), version 2.0: Imaging software to extract canopy structure and gap light transmission indices from true-colour fisheye photographs, users manual and program documentation. Simon Fraser University, Burnaby, British Columbia; and the Institute of Ecosystem Studies, Millbrook, New York, USA.
- GÓMEZ, A., AND G. R. CARVALHO. 2000. Sex, parthenogenesis and genetic structure of rotifers: Microsatellite analysis of contemporary and resting egg bank populations. *Molecular Ecology* 9: 203–214.
- HELD, M. E. 1983. Pattern of beech regeneration in the east-central United States. *Bulletin of the Torrey Botanical Club* 110: 55–62.
- HOLSINGER, K. E. 2000. Reproductive systems and evolution in vascular plants. *Proceedings of the National Academy of Sciences* 97: 7037–7042.
- IVERSON, L. R., A. M. PRASAD, S. N. MATTHEWS, AND M. PETERS. 2008. Estimating potential habitat for 134 eastern US tree species under six climate scenarios. *Forest Ecology and Management* 254: 390–406.
- JONES, R. H., AND D. J. RAYNAL. 1986. Spatial distribution and development of root sprouts in *Fagus grandifolia* (Fagaceae). *American Journal of Botany* 73: 1723–1731.

- JONES, R. H., AND D. J. RAYNAL. 1988. Root sprouting in American beech (*Fagus grandifolia*): Effects of root injury, root exposure, and season. *Forest Ecology and Management* 25: 79–90.
- KITAMURA, K., K. HOMMA, H. TAKASU, S. HAGIWARA, F. UTECH, D. F. WINGHAM, AND S. KAWANO. 2001. Demographic genetics of the American beech, *Fagus grandifolia*. II. Genet substructure of populations for the Blue Ridge, Piedmont and the Great Smoky Mountains. *Plant Species Biology* 16: 219–230.
- KITAMURA, K., AND S. KAWANO. 2001. Regional differentiation in genetic components for the American beech, *Fagus grandifolia* Ehrh., in relation to geological history and mode of reproduction. *Journal of Plant Research* 114: 353–368.
- KOCH, J. L. 2010. Beech bark disease: The oldest “new” threat to American beech in the United States. *Outlooks on Pest Management* 21: 64–68.
- KUBISIAK, T., D. W. CAREY, C. BURDINE, AND J. KOCH. 2009. Characterization of ten EST-based microsatellite loci isolated from American beech, *Fagus grandifolia* Ehrh. *Molecular Ecology Resources* 9: 1460–1466.
- MASON, M. E., J. L. KOCH, M. KRASOWSKI, AND J. LOO. 2013. Comparisons of protein profiles of beech bark disease resistant and susceptible American beech (*Fagus grandifolia*). *Proteome Science* 11: 2.
- MATTHEWS, S. N., L. R. IVERSON, A. M. PRASAD, M. P. PETERS, AND P. G. RODEWALD. 2011. Modifying climate change habitat models using tree species-specific assessments of model uncertainty and life history-factors. *Forest Ecology and Management* 262: 1460–1472.
- MEIRMANS, P. G., AND P. H. VAN TIENDEREN. 2004. GENOTYPE and GENODIVE: Two programs for the analysis of genetic diversity of asexual organisms. *Molecular Ecology Notes* 4: 792–794.
- MORRIS, A. B., R. L. SMALL, AND M. B. CRUZAN. 2004. Variation in frequency of clonal reproduction among populations of *Fagus grandifolia* Ehrh. in response to disturbance. *Castanea* 69: 38–51.
- PIELOU, E. 1969. An introduction to mathematical ecology. Wiley-Interscience, New York, New York, USA.
- PYLE, C. 1988. The type and extent of anthropogenic vegetation disturbance in the Great Smoky Mountains before National Park Service acquisition. *Castanea* 53: 183–196.
- RUSSELL, N. H. 1953. The beech gaps of the Great Smoky Mountains. *Ecology* 34: 366–374.
- SILVERTOWN, J. 2008. The evolutionary maintenance of sexual reproduction: Evidence from the ecological distribution of asexual reproduction in clonal plants. *International Journal of Plant Sciences* 169: 157–168.
- TAKAHASHI, K., K. ARII, AND M. J. LECHOWICZ. 2010. Codominance of *Acer saccharum* and *Fagus grandifolia*: The role of *Fagus* root sprouts along a slope gradient in an old-growth forest. *Journal of Plant Research* 123: 665–674.
- TUBBS, C. H., AND D. R. HOUSTON. 1990. *Fagus grandifolia* Ehrh. American beech. In R. M. Burns and B. H. Honkala [eds.], *Silvics of North America: 2. Hardwoods*, vol. 2. U.S. Department of Agriculture, Forest Service, Washington, D.C., USA.
- VALLEJO-MARIN, M., M. E. DORKEN, AND S. C. H. BARRETT. 2010. The ecological and evolutionary consequences of clonality for plant mating. *Annual Review of Ecology, Evolution and Systematics* 41: 193–213.
- VAN GROENENDAEL, J. K., L., KLIMES, J. KLIMESOVA, AND R. HENDRIKS. 1997. Comparative ecology of clonal plants. *Philosophical Transactions of the Royal Society of London, B, Biological Sciences* 351: 1331–1339.
- WARD, R. T. 1961. Some aspects of the regeneration habits of the American beech. *Ecology* 42: 828–832.
- WHITTAKER, R. H. 1956. Vegetation of the Great Smoky Mountains. *Ecological Monographs* 26: 1–80.
- WIGGINS, G. J., J. F. GRANT, M. T. WINDHAM, R. A. VANCE, B. RUTHERFORD, R. KLEIN, K. JOHNSON, ET AL. 2004. Associations between causal agents of the beech bark disease complex [*Cryptococcus fagisuga* (Homoptera: Cryptococcidae) and *Nectria* spp.] in the Great Smoky Mountains National Park. *Environmental Entomology* 33: 1274–1281.

APPENDIX 1. Population voucher specimens for sites included in the current study.

Taxon; Sampling site code; *Voucher specimen*, Collection locale; Herbarium.

Fagus grandifolia* Ehrh.;** GRB; AM301, AM317, AM318, Great Smoky Mountains National Park (GRSM), Tennessee; MTSU. ***F. grandifolia* Ehrh.;** HR; AM352, AM361, AM370, GRSM, North Carolina; MTSU. ***F.

***grandifolia* Ehrh.;** CHM; AM379, AM389, AM390, GRSM, Tennessee; MTSU. ***F. grandifolia* Ehrh.;** BMT; AM331, AM335, AM342, AM343, GRSM, North Carolina; MTSU.